

EXPERIMENT NO. 9:

Compare Linear and Polynomial Regression using Python

Aim:

To compare the performance of **Linear Regression** and **Polynomial Regression** using Python on a sample dataset.

Procedure:

1. Import the required Python libraries.
2. Create a sample dataset.
3. Train a Linear Regression model.
4. Transform the data into polynomial features.
5. Train a Polynomial Regression model.
6. Predict the output using both models.
7. Compare the R^2 scores of both models.
8. Display the predicted values and performance comparison.

Program:

```
# Compare Linear and Polynomial Regression
```

```
import numpy as np

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures

from sklearn.metrics import r2_score
```

```
# Sample Dataset
```

```
X = np.array([[1],[2],[3],[4],[5],[6],[7],[8],[9],[10]])

y = np.array([1,4,9,16,25,36,49,64,81,100])
```

```
# Linear Regression
```

```
linear_model = LinearRegression()

linear_model.fit(X, y)

linear_pred = linear_model.predict(X)
```

```
# Polynomial Regression

poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = LinearRegression()
poly_model.fit(X_poly, y)
poly_pred = poly_model.predict(X_poly)

# R2 Scores

linear_score = r2_score(y, linear_pred)
poly_score = r2_score(y, poly_pred)

print("Linear Regression R2 Score:", round(linear_score, 2))
print("Polynomial Regression R2 Score:", round(poly_score, 2))
```

Output:

```
... Linear Regression R2 Score: 0.95
    Polynomial Regression R2 Score: 1.0
```

Result:

The **Linear Regression** and **Polynomial Regression** models were successfully implemented and compared. The **Polynomial Regression** model achieved a higher **R² score (1.0)** than the **Linear Regression** model, indicating a better fit for the given nonlinear dataset.